



Welcome to AWS Academy Cloud Architecting

AWS Academy Cloud
Architecting

Module objectives

This module prepares you to do the following:

- Recognize the basic elements of the café business case.
- Describe the role of a cloud architect.

Module overview

Presentation sections

- Course overview
- Café business case introduction
- Roles in cloud computing



Course overview

Welcome to AWS Academy Cloud Architecting

Course objectives

- Apply Amazon Web Services (AWS) architectural principles and best practices to make architectural decisions.
- Use appropriate AWS services and features to make infrastructure scalable, reliable, and highly available.
- Choose AWS managed services to enable greater flexibility and resiliency in an infrastructure.
- Increase performance and reduce cost of a cloud infrastructure built on AWS.
- Use AWS services and features to secure user, application, and data access.
- Apply best practices from the AWS Well-Architected Framework to improve architectures that use AWS solutions.

Alignment to AWS Certification

The concepts in this course align to the knowledge and skills described in the AWS Certified Solutions Architect – Associate exam (SAA-C03) guide.

The following knowledge and skills are validated by the AWS Certified Solutions Architect exam:

- Ability to design solutions based on the AWS Well-Architected Framework
- Ability to perform the following tasks:
 - Design solutions that incorporate AWS services to meet current business requirements and future projected needs.
 - Design architectures that are secure, resilient, high-performing, and cost-optimized.
 - Review existing solutions and determine improvements.



Course outline by module

1. Welcome to AWS Academy Cloud Architecting
2. Introducing Cloud Architecting
3. Securing Access
4. Adding a Storage Layer with Amazon S3
5. Adding a Compute Layer Using Amazon EC2
6. Adding a Database Layer
7. Creating a Networking Environment
8. Connecting Networks
9. Securing User, Application, and Data Access
10. Implementing Monitoring, Elasticity, and High Availability
11. Automating Your Architecture
12. Caching Content
13. Building Decoupled Architectures
14. Building Serverless Architectures and Microservices
15. Data Engineering Patterns
16. Planning for Disaster
17. Bridging to Certification

Suggested starting point for this course

- You will be most successful in this course if you have the following:
 - Knowledge of AWS Cloud topics equivalent with completing the AWS Academy Cloud Foundations or a similar course
 - Working knowledge of distributed systems
 - Working knowledge of multi-tier architectures
 - Familiarity with general networking concepts
- Find self-paced training links for these topics on the content resources page of your course.



Course materials

Learning materials

- Student guides and videos based on the instructor presentations for each module
- Hands-on labs include the following:
 - Guided labs (step-by-step instructions)
 - Challenge labs (based on a business case)
 - AWS Academy Cloud Architecting Capstone
- Recorded demonstrations
- Instructor-led activities

Knowledge checks

- A 10-question knowledge check for each module, from Modules 2 through 16
- A sample certification exam question in the student guide for each module, from Modules 2 through 16
- A 25-question course assessment drawn from slides and student guides for all modules



Café business case introduction

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Café business scenario

- Frank and Martha opened a café and bakery in their retirement.
- Their staff and a few AWS consultants who are also customers are helping them use the cloud to address the needs of their growing business.



Café business owners and staff



Frank

- Co-owner of the café
- Retired from the navy
- Likes to bake
- Nontechnical



Martha

- Co-owner of café
- Retired accountant
- Knows how to use spreadsheets but otherwise, nontechnical



Sofía

- Daughter of Frank and Martha
- Supply chain manager for the café
- Future business administration student
- Has technical skills that include programming



Nikhil

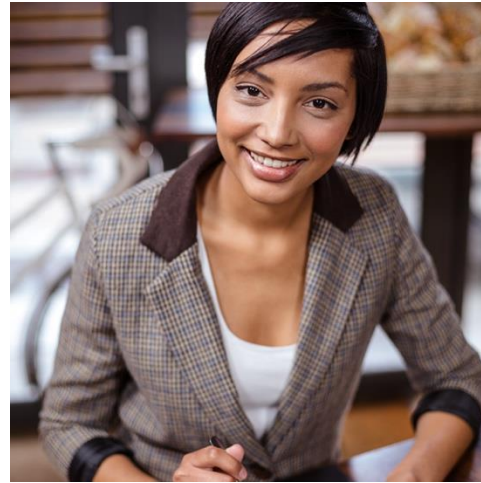
- Café employee with visual design skills
- Interested in learning cloud computing
- Might take on more responsibilities at the café when Sofía starts her university studies

Café visitors who are AWS consultants



Olivia

- AWS solutions architect
- Has technical skills that include a specialty in databases and network technologies



Faythe

- Developer
- Has experience with AWS programming interfaces
- Is knowledgeable about cloud security



Mateo

- Systems administrator and engineer
- Likes to find ways to automate and to create repeatable solutions
- Knows the importance of backups and disaster recovery in solution design

The evolving café architecture

Version	Business Reason for Update	Technical Requirements and Architecture Update
V1	Build a static website for a small business.	Host the website on Amazon S3.
V2	Update the website to support dynamic content and online ordering.	Deploy the web application and database on Amazon EC2.
V3	Reduce the effort to maintain the database and secure its data.	Separate web and database layers. Migrate the database to Amazon RDS on a private subnet.
V4	Enhance the security of the web application.	Use Amazon VPC features to configure and secure public and private subnets.
V5	Ensure that the website can handle an expected increase in traffic and remain highly available and resilient to failure.	Add a load balancer, implement auto scaling on the EC2 instances, and distribute compute and database instances across two Availability Zones.
V6	Automate deployments so that the café can consistently deploy, manage, and update café resources across Regions.	Build a version-controlled CloudFormation template to deploy the network and application layers. Deploy the CloudFormation stack to another Region.
V7	Add reporting capabilities while reducing the operational and maintenance burden, improving performance, and reducing costs.	Deploy Lambda functions that connect to the Amazon RDS database and generate a report based on a schedule.





Roles in cloud computing

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IT professional



Common skills and responsibilities include the following:

- Act as a generalist and might manage an application
- Often manage a production environment
- Highly technical
- Might have significant or limited experience in cloud technologies
- Might specialize in one area (such as security or storage)

Job titles include the following:

- IT administrator
- Systems administrator
- Network administrator

IT leader



Common skills and responsibilities include the following:

- Lead a team of IT professionals
- Responsible for day-to-day operations
- Manage a budget, stay informed about technologies, and choose new technologies
- Hands on during the early stages of a project, and then delegate the team to take over

Job titles include the following:

- IT manager
- IT director
- IT supervisor

Developer



Common skills and responsibilities include the following:

- Write, test, and fix code
- Think about projects at the application level
- Work with APIs and SDKs
- Might use sample code
- Might specialize in one area (such as security or storage)

Job titles include the following:

- Software developer
- System architect
- Software development manager

DevOps engineer



Common skills and responsibilities include the following:

- Build out the infrastructure that applications run on, often in the cloud
- Follow the guidelines of the cloud architect
- Experiment to improve deployments

Job titles include the following:

- DevOps engineer
- Build engineer
- Reliability engineer

Cloud architect



Common skills and responsibilities include the following:

- Stay up to date with new technologies and help decide which ones to use
- Provide documentation, processes, and tooling to developers
- Give developers freedom to innovate
- Resolve challenges using best practices for cost optimization, performance, reliability, and security

Job titles include the following:

- Cloud architect
- Systems engineer
- Systems analyst



Module wrap-up

Welcome to AWS Academy Cloud Architecting

Module summary

This module prepared you to do the following:

- Recognize the basic elements of the café business case.
- Describe the role of a cloud architect.



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.